PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review
AUTHORS	Dahl, Katrine; Andersen, Mads; Henriksen, Tine

VERSION 1 – REVIEW

REVIEWER	Garstang, Joanna University of Birmingham, College of Medical and Dental Sciences
REVIEW RETURNED	14-Aug-2021

GENERAL COMMENTS	This is an interesting and well written systematic review that should be worthy of publication but would be improved by some revisions. It appears to lack a search of grey literature which is a major weakness given that negative reporting studies may remain unpublished. Given the very weak association (if any) between NMSIDS and SIDS I am not sure that including this as a secondary outcome is helpful, particularly as all these papers are approaching 40 years old. The discussion section would benefit from editing and following a clearer structure. I have made a few suggestions for minor amendments: Introduction Pg 5 Line 17 – saying there is 'no explanation' for bereaved parents is a bit extreme. There is considerable knowledge of risk factors & the triple risk model – this is how doctors may explain SIDS to parents. Pg 5 Line 20 – it would be better to state there is limited evidence that near-miss SIDS is a precursor to SIDS. Methods Pg 6 line 23– is google translate really adequate for a systematic review? Should a formal translation be obtained if a paper is to be included? How many papers were included with a google translation? Were there any limits by date of publication or location? Pg 6 line 49. The San Diego definition of SIDS was published in 2004. How was SIDS defined in papers published before this date? How did you account for studies which may have labelled cases as SIDS without adequate investigation (eg post-mortem ancillary testing, histology, metabolics, infection, medical history, death scene analysis etc) Pg 7 line 6 – did you do hand searches, grey literature, contact academics, and snow-ball searches? This is a requirement for a systematic review. Results Pg 9 line 12. ALTE is considered the same as NMSIDS, it is the preferred UK term for example – so you did include studies where this was the outcome, it was just named differently.

	Table 1 – did Orlowski and Stockard separate the results from NMSIDS and SIDS cases, or analyse these jointly? Pg 13 line 6 – didn't Orlowski and Stockard include both NMSIDS and SIDS cases? That is what is stated in table 1. This should be clarified. Table 2 what does (*) mean? Pg 17 line 10 – I'm not sure what you mean by 'NMSIDS as a surrogate outcome for SIDS were downgraded for indirectness'. Discussion The first 2 pages of discussion is actually better placed in the results section, as it is just describing the findings. The discussion would be easier to follow if it used the standard BMJ format: Statement of principal findings Strengths & weaknesses of the study Strengths & weaknesses in relation to other studies & key differences Possible mechanisms & explanations for findings Potential implications for clinicians or policymakers Unanswered questions and future research

REVIEWER	Kileny, Paul R
REVIEW RETURNED	19-Aug-2021

GENERAL COMMENTS	This is a narrative, systematic review of human studies related to the association of auditory system anomalies with SIDS. The authors reviewed functional (ABR, OAE), and anatomical/histological studies for the purpose of this analysis. The definition of bias is not very clear, especially if one is familiar with the studies cited here. For instance, 42 and 43 are both listed as presenting a high risk of bias, in spite of the fact that the methods, and controls were completely different - one (42) with carefully selected, prospective controls. Moreover, 43, 60 , and 61, actually use the very same data - these authors essentially published the same study three times, and this was a known fact at the time. 45 (Stockard) is not a peer reviewed publication, so I am not sure it belongs to this review. Another deficit of this review, is that it lacks any assessment of the validity of the methods used to record and evaluate electrophysiological events related to auditory function. Unless at least one of the authors has that type of expertise, simply doing a publication search cannot accomplish that. In other words, poorly executed studies are at times published. The authors should address these issues.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Comment 1
This is an interesting and well written systematic review that should be worthy of publication but would be improved by some revisions. It appears to lack a search of grey literature which is a major weakness given that negative reporting studies may remain unpublished. Given the very weak association (if any) between NMSIDS and SIDS I am not sure that including this as a secondary outcome is helpful, particularly as all these papers are approaching 40 years old. The discussion section would benefit from editing and following a clearer structure.

Reply
Thank you for your thoughtful review of our manuscript. We have taken all your comments into consideration and feel that the manuscript has improved accordingly.

As suggested, we have now searched for grey literature according to the procedure described by the Cochrane Handbook and added the following paragraph in the manuscript:
“Additionally, we hand-searched grey literature by the use of Open Grey, Ovid, and The National Technical Information Service without any additional result.(28)”

We agree that the association between SIDS and NMSIDS may be weak. We have chosen to compile all current evidence within this narrow subject: the association between auditory system pathology and SIDS, which included NMSIDS. We are aware of the difficulties that this may cause and have clarified and further discussed this in the Discussion section.

The following paragraph is added to the Discussion section: “Additionally, there may be some concern regarding the evidence from the NMSIDS studies. Firstly, we assessed the studies to have a higher degree of risk of bias compared to the SIDS studies. Secondly, the NMSIDS studies are from the 1980s when the SIDS incidence was higher and the knowledge on risk factors associated with SIDS were sparse. Therefore, the generalizability to more recent times may be questionable due to differences in patient populations. Thirdly, a NMSIDS event may not be associated with SIDS.(5-7) Actually, the objectives in the NMSIDS studies were to examine ABR as a tool to examine the brainstem in general rather than the auditory function. By these concerns, the NMSIDS studies were of especially poor quality and did not influence our overall conclusion.

Comment 2
Pg 5 Line 17 – saying there is ‘no explanation’ for bereaved parents is a bit extreme. There is considerable knowledge of risk factors & the triple risk model – this is how doctors may explain SIDS to parents.

Reply
We acknowledge that we have come a long way in the understanding of SIDS by exploring risk factors and the triple risk model.

The paragraph previously read: “However, SIDS is still one of the leading causes of death in infancy (0.30 per 1,000 live births) and a scientific mystery with no explanation for the bereaved parents.(4)”

It now reads: “However, SIDS is still one of the leading causes of death in infancy (0.30 per 1,000 live births) and a scientific mystery.(4)”

Comment 3
Pg 5 Line 20 – it would be better to state there is limited evidence that near-miss SIDS is a precursor to SIDS.

Reply
The paragraph on this previously stated: “A near-miss SIDS (NMSIDS) episode may be considered as a precursor before an actual SIDS event. However, the link to SIDS is discussed vividly.(5-7) NMSIDS is also classified as an apparent life-threatening event (ALTE), which to some degree have replaced the term NMSIDS.(7)”

It now reads: “A near-miss SIDS (NMSIDS) episode may be considered as a precursor before an actual SIDS event. However, the evidence for this association is limited.(5-7) NMSIDS is also known as an apparent life-threatening event (ALTE), which to some degree have replaced the term

NMSIDS.(7)"

Comment 4
Pg 6 line 23– is google translate really adequate for a systematic review? Should a formal translation be obtained if a paper is to be included? How many papers were included with a google translation? Were there any limits by date of publication or location?

Reply
We definitely agree that the use of Google Translate has translational limitations. However, it was adequate during the full-text screening to exclude three studies in other languages than English – and no other studies in other languages than English were included.

We have added the following paragraph to the Results section: "No studies in other languages than English were included and no additional studies were identified by snowball search or search for grey literature."

The following paragraph has been added in the Method section: "No limitations regarding publication date or location were implemented."

Comment 5
Pg 6 line 49. The San Diego definition of SIDS was published in 2004. How was SIDS defined in papers published before this date? How did you account for studies which may have labelled cases as SIDS without adequate investigation (eg post-mortem ancillary testing, histology, metabolics, infection, medical history, death scene analysis etc)

Reply
In Table 1, we have compiled the different SIDS and NMSIDS definitions used in the included studies. In the initial planning phase, we considered only including studies with a clear definition of SIDS; however, we wanted to examine the current evidence in large. In the discussion, we have clarified the limitations regarding this approach. Additionally, an inadequate definition of SIDS/NMSIDS led to downgrading by the Newcastle-Ottawa Scale during the risk of bias assessment. Furthermore, studies were excluded if cases were above 1 year of age.

The following paragraph is added to the Discussion section: "We included three different exposures: OAE, ABR and brainstem histology of the auditory system and two different outcomes: SIDS and NMSIDS, which were either rather ill-defined or prone to different definitions between studies. Only one study (37) used the San Diego definition of SIDS (1). The differences in exposure and outcome definition comprised comparability between studies and in case of inadequate outcome definition, the quality of the individual study was downgraded during the NOS assessment."

Comment 6
Pg 7 line 6 – did you do hand searches, grey literature, contact academics, and snow-ball searches? This is a requirement for a systematic review.

Reply
Grey literature has now been hand-searched without any additional results. Regarding snowball searches, references of included studies were scrutinized as well as citations identified in Scopus.

The following paragraph is found in the Method section: "References of included studies were scrutinized as well as citations identified in Scopus."

The following paragraph has been added to the Result section: "No studies in other languages than

English were eligible for inclusion and no additional studies were identified by snowball search or search for grey literature."

Please, see Reply to Reviewer 1, Comment 1.

Comment 7
Pg 9 line 12. ALTE is considered the same as NMSIDS, it is the preferred UK term for example – so you did include studies where this was the outcome, it was just named differently.

Replay
The paragraph previously stated following in the Methods section: "We included NMSIDS and ALTE as secondary outcomes. NMSIDS was defined as severe life-threatening apnoea, while ALTE was defined as a combination of apnoea, marked changes in muscle tone, colour change, choking or gasping, and an apparent need for resuscitation by vigorous stimulation or ventilation.(9, 27)"

In accordance with the comment above, it now reads: "We included NMSIDS and ALTE as secondary outcomes. NMSIDS and ALTE were defined as severe life-threatening apnoea with marked changes in muscle tone, colour change, choking or gasping, and an apparent need for resuscitation by vigorous stimulation or ventilation.(9, 27)"

Comment 8
Table 1 – did Orlowski and Stockard separate the results from NMSIDS and SIDS cases, or analyse these jointly?

Reply
Orlowski and Stockard only included NMSDIS cases; however, one case in Stockard's publication eventually died of SIDS. Nonetheless, both studies included a definition of SIDS. As no SIDS cases were identified, we have deleted their definitions in Table 1 to avoid any future confusion.

Comment 9
Pg 13 line 6 – didn't Orlowski and Stockard include both NMSIDS and SIDS cases? That is what is stated in table 1. This should be clarified.

Reply
Please refer to the comment above.

Comment 10
Table 2 what does (*) mean?

Reply
According to the Newcastle-Ottawa Scale, each study could be awarded a maximum of one star/point (*) for each numbered item within the Selection and Exposure categories and a maximum of two stars/points (*) for Comparability.

The explanation of Table 2 previously stated: "a A) yes, with independent validation; B) yes, e.g. record linkage or based on self-reports; C) no description. b A) consecutive or obviously representative series of cases; B) potential for selection biases or not stated. c A) community controls; B) hospital controls; C) no description. d A) no history of endpoint; B) no description of source. e A) study controls for sleeping position and conditions; B) study controls for sex, smoking exposure, birth weight or gestational age. f A) secure record (e.g. surgical records); B) structured interviews were blind to case/control status; C) written self-report or medical record only; D) no description. g A) yes; B) no. h A) same rate for both groups; B) non-respondents described; C) rate different and no

designation. l “low risk” was given 3-4 points in selection and 1-2 points in comparability and 2-3 points in outcome; “fair risk” was given 2 points in selection and 1-2 points in comparability and 2-3 points in outcome; “high risk’ was given 0-1 point in selection or 0 points in comparability or 0-1 point in outcome”

It now reads: “a A) yes, with independent validation (*); B) yes, e.g. record linkage or based on self-reports; C) no description. b A) consecutive or obviously representative series of cases (*); B) potential for selection biases or not stated. c A) community controls (*); B) hospital controls; C) no description. d A) no history of endpoint (*); B) no description of source. e A) study controls for sleeping position and conditions (*); B) study controls for sex, smoking exposure, birth weight or gestational age (*). f A) secure record (e.g. surgical records) (*); B) structured interviews were blind to case/control status (*); C) written self-report or medical record only; D) no description. g A) yes (*); B) no. h A) same rate for both groups (*); B) non-respondents described; C) rate different and no designation. l “low risk” was given 3-4 points in selection and 1-2 points in comparability and 2-3 points in outcome; “fair risk” was given 2 points in selection and 1-2 points in comparability and 2-3 points in outcome; “high risk’ was given 0-1 point in selection or 0 points in comparability or 0-1 point in outcome.”

Furthermore the (*) is now explained in the Methods section of the manuscript: “A maximum of nine points (*) could be awarded to each study by NOS. “Low” risk of bias required 3-4 points in selection, 1-2 points in comparability, and 2-3 points in exposure; “fair” risk required 2 points in selection, 1–2 points in comparability, and 2–3 points in exposure; and ‘high” risk was given 0-1 point in selection or 0 points for comparability or 0-1 point in exposure. Regarding comparability, one point was awarded if the study adjusted or matched for sleeping position or condition, while an additional point was awarded if they adjusted for either maternal smoking, sex, gestational age at birth, or birth weight.”

Comment 11
Pg 17 line 10 – I’m not sure what you mean by ‘NMSIDS as a surrogate outcome for SIDS were downgraded for indirectness’.

Reply
As NMSIDS may not be associated with SIDS (please also refer to replies above), we downgraded NMSIDS studies to emphasize this when the quality of evidence (GRADE) was assessed.

The paragraph previously stated: “Furthermore, NMSIDS as a surrogate outcome for SIDS were downgraded for indirectness.”

It now reads: “Furthermore, NMSIDS as a surrogate outcome for SIDS were downgraded for indirectness according to the GRADE assessment, as it is unclear whether NMSIDS is in fact associated with SIDS.
Comment 12
The first 2 pages of discussion is actually better placed in the results section, as it is just describing the findings.
The discussion would be easier to follow if it used the standard BMJ format:
Statement of principal findings
Strengths & weaknesses of the study
Strengths & weaknesses in relation to other studies & key differences
Possible mechanisms & explanations for findings
Potential implications for clinicians or policymakers
Unanswered questions and future research

Reply

Thank you for your advice concerning the discussion, which has now been revised. Among other things, we have now added subject headings to create a better overview (please refer to the Discussion section in the revised manuscript). We have added elements of the above format to fit our systematic review.

Reviewer 2
Comment 1
This is a narrative, systematic review of human studies related to the association of auditory system anomalies with SIDS. The authors reviewed functional (ABR, OAE), and anatomical/histological studies for the purpose of this analysis.
The definition of bias is not very clear, especially if one is familiar with the studies cited here. For instance, 42 and 43 are both listed as presenting a high risk of bias, in spite of the fact that the methods, and controls were completely different - one (42) with carefully selected, prospective controls. Moreover, 43, 60, and 61, actually use the very same data - these authors essentially published the same study three times, and this was a known fact at the time. 45 (Stockard) is not a peer reviewed publication, so I am not sure it belongs to this review.
Another deficit of this review, is that it lacks any assessment of the validity of the methods used to record and evaluate electrophysiological events related to auditory function. Unless at least one of the authors has that type of expertise, simply doing a publication search cannot accomplish that. In other words, poorly executed studies are at times published. The authors should address these issues.

Reply
Thank you for your thoughtful review of our manuscript. We have taken all your comments into consideration and feel that the manuscript has improved accordingly.

The risk of bias was assessed based on what information was available in the studies. Table 2 provides an overview of the risk of bias assessment and in which area one may have concerns. By using the Newcastle-Ottawa scale, the risk of bias assessment is attempted to be standardized. Studies with a clearly defined control group may still be assessed as having a high risk of bias, as they may be compromised in other areas e.g., selection of cases.

We agree with your concerns regarding references 43, 60, and 61. We have opted to now only mention the study of Orlowski (44) to avoid indicating a stronger association.

To our knowledge, the study by Stockard is peer-reviewed; it has been published in a peer-reviewed journal (Annals of the New York Academy of Sciences). In our protocol, we did not specify that only peer-reviewed studies were eligible for inclusion. We aimed to compile all information regarding this subject. Nonetheless, we definitely agree that this comes with some limitation.

Two out of three authors have worked with these measures during experimental research. However, we did not have a decidedly specialist to examine the OAE and ABR records. This could definitely have strengthened our systematic review. This concern has now been addressed in the Discussion section.

The following paragraph has been added to the Discussion section: “We had no expert evaluation of the OAE and ABR measures used in the included studies. Consequently, we may have overrated the quality of these.”

VERSION 2 – REVIEW

REVIEWER	Garstang, Joanna University of Birmingham, College of Medical and Dental Sciences
REVIEW RETURNED	04-Oct-2021

GENERAL COMMENTS	The revisions have addressed all the issues raised previously, and I think it is a useful contribution to the scientific literature on SIDS. I have one minor comment - you cannot confirm SIDS on autopsy. It is more accurate to state SIDS diagnosis was given following the absence of any other pathological diagnosis at autopsy.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Comment 1
The revisions have addressed all the issues raised previously, and I think it is a useful contribution to the scientific literature on SIDS.
I have one minor comment - you cannot confirm SIDS on autopsy. It is more accurate to state SIDS diagnosis was given following the absence of any other pathological diagnosis at autopsy.

Reply
Thank you for your thoughtful review of our manuscript. We have taken your comment into consideration and feel that the manuscript has improved accordingly.

As suggested following alterations have been made.

In the Introduction section:
The paragraph previously read: “Sudden Infant Death Syndrome (SIDS) is defined as the sudden death of an infant under the age of one year occurring unexpectedly, with no other potential explanation of death and an autopsy compatible with SIDS.(1)”

It now reads: “Sudden Infant Death Syndrome (SIDS) is defined as the sudden death of an infant under the age of one year occurring unexpectedly, with no other potential explanation of death and absence of any other pathological diagnosis at autopsy.(1)”

In the Results section:
The paragraph previously read: “All SIDS diagnosis was confirmed by autopsy in all infants.”

It now reads: “All SIDS diagnoses were given following the absence of any other pathological diagnosis at autopsy.”